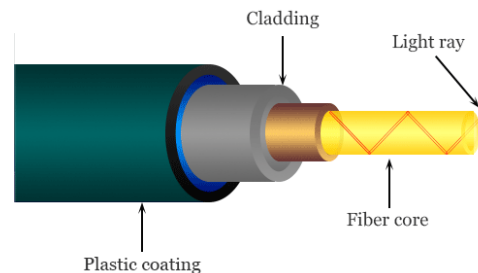


1. What is Optical Fiber?

Solution:

Optical fiber is a flexible, transparent material consisting of a core and cladding, which is used for the transmission of light rays.

The light is restricted within the core due to the difference in refractive indices between the core and the cladding, allowing it to travel long distances with minimal signal loss.



2. Describe the electromagnetic spectrum showing the region used for optical fiber communication?

Solution:

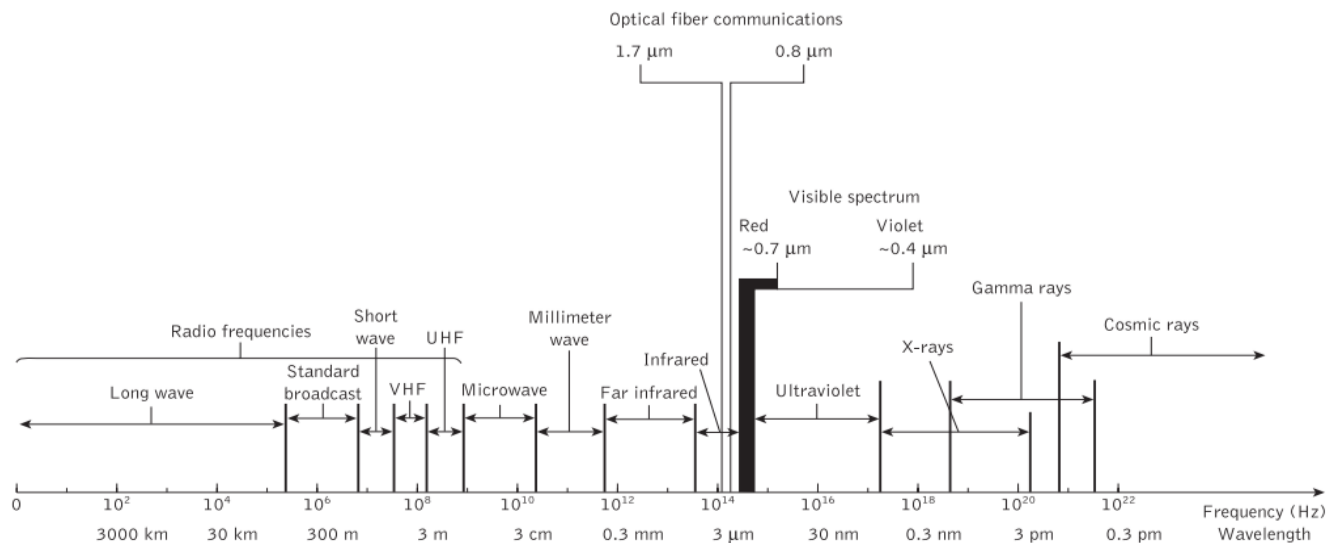


Figure 1.1 The electromagnetic spectrum showing the region used for optical fiber communications

The diagram shows the electromagnetic spectrum, ranging from low-frequency radio waves to high-frequency gamma rays. It highlights the optical fiber communication region, which uses infrared light with wavelengths between 0.8 μm and 1.7 μm . This range is ideal for data transmission due to low signal loss. The visible spectrum (0.4 μm to 0.7 μm) is also shown, where human vision operates, but optical fiber communication operates just beyond the visible red light, in the infrared region.

3. What are the difference between the overall communication system and the optical fiber communication system?

Solution:

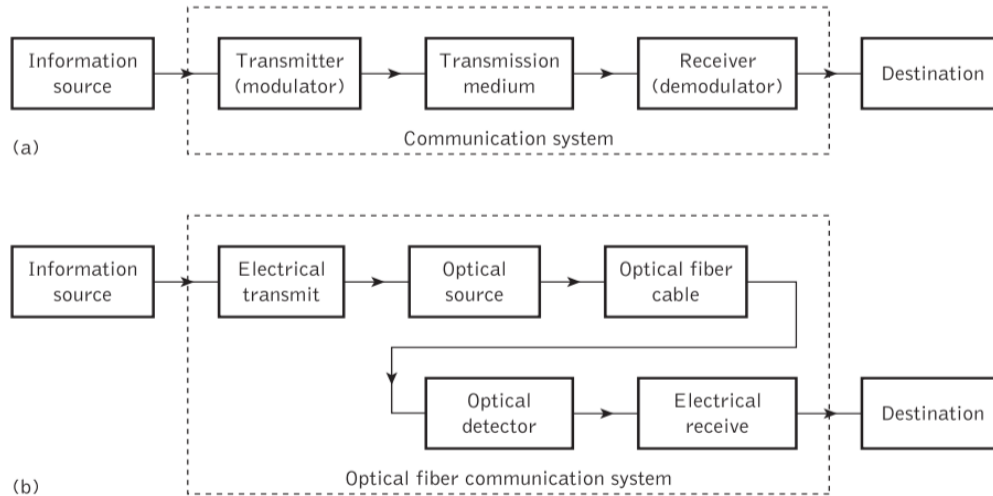


Figure 1.2 (a) The general communication system. (b) The optical fiber communication system

Aspect	Optical Fiber Communication	General Communication
Transmission Medium	Optical fibers	Wires (twisted pair cable, coaxial cable) or wireless (radio waves)
Carrier/Modulation	Light waves	Electromagnetic waves (radio, microwave)
Bandwidth	Extremely high (capable of terabit per second speeds)	Lower, varies depending on the medium (e.g., radio, copper)
Maximum Distance	Long distances without repeaters	Shorter maximum distance, requires repeaters/amplifiers
Repeaters/Amplifiers	Installed at long intervals due to low signal loss	More frequent need due to higher signal loss
Modulation Technique	Analog (continuous) and digital (on-off pulses)	Analog (AM, FM) and digital (ASK, FSK, PSK)
Signal Attenuation	Low attenuation over long distances	High attenuation over distance
Noise and Interference	Immune to electromagnetic interference	Affected by electromagnetic interference.
Signal Distortion	Minimal signal distortion due to dispersion and attenuation	Distortion due to electromagnetic interference
Efficiency	Higher.	Lower.
Implementation Cost	Higher initial cost, lower maintenance	Lower initial cost, higher maintenance.
Usage	Internet backbone, data centers, high-speed networks	Residential broadband, TV, mobile communication

4. Explain fiber optic communication system with a suitable block diagram?

Solution:

The optical carrier may be modulated using either an analog or digital information signal. In the system analog modulation involves the variation of the light emitted from the optical source in a continuous manner. With digital modulation, however, discrete changes in the light intensity are obtained (i.e. on-off pulses).

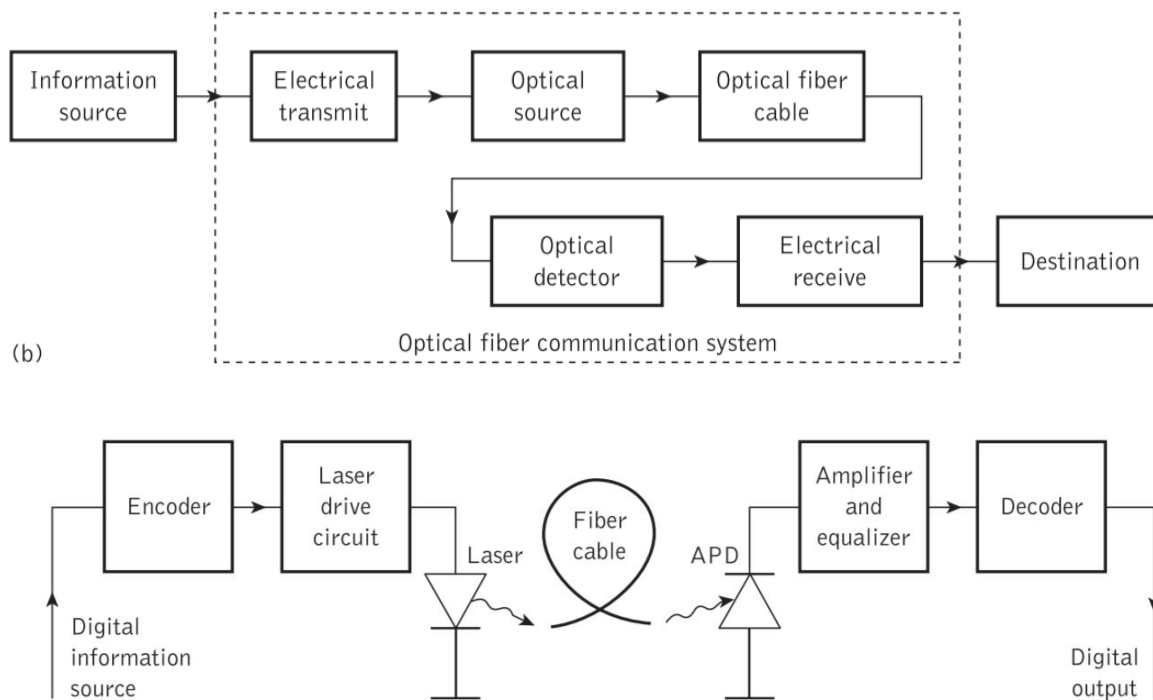


Figure 1.3 A digital optical fiber link using a semiconductor laser source and an avalanche photodiode (APD) detector

- Initially, the **input digital signal** from the information source is suitably **encoded** for optical transmission.
- The **laser drive circuit** directly modulates the intensity of the semiconductor laser with the encoded digital signal.
- Hence a digital optical signal is launched into the optical fiber cable.
- The Avalanche Photodiode (APD) detector is followed by a front-end amplifier and equalizer or filter to provide
 - gain as well as linear signal processing and
 - noise bandwidth reduction
- Finally, the signal obtained is decoded to give the original digital information.

5. Typical fiber optic configuration with required schema? - physical setup

Solution:

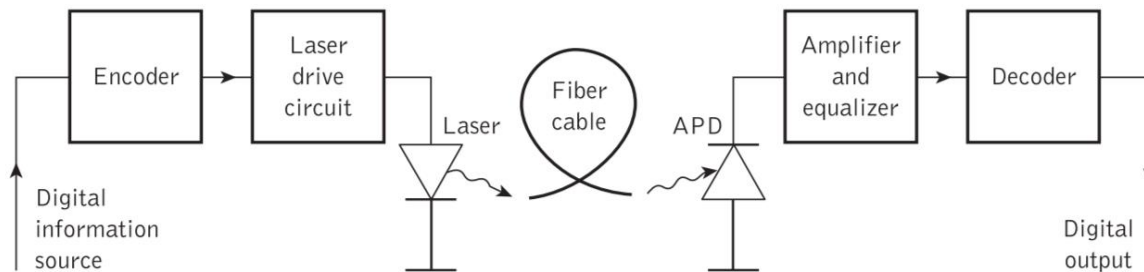


Figure 1.3 A digital optical fiber link using a semiconductor laser source and an avalanche photodiode (APD) detector

A **typical fiber optic communication system** consists of several key components working together to transmit information via light signals through optical fibers. Below is the configuration along with a basic explanation of each component:

Components of a Typical Fiber Optic Communication System:

1. **Information Source:** The initial source of data, which can be in the form of audio, video, or digital data.
2. **Transmitter:**
 - **Electrical Signal Stage:** Converts the information source into an electrical signal.
 - **Optical Source:** Converts the electrical signal into a light signal. This could be a Laser Diode (LD) or a Light Emitting Diode (LED).
 - **Modulation:** The light signal is modulated to carry the information.
 -
3. **Optical Fiber (Transmission Medium):** The optical fiber, typically made of glass or plastic, transmits the modulated light signal over long distances with minimal loss. It consists of:
 - **Core:** The central region where light travels.
 - **Cladding:** Surrounds the core and keeps the light within the core by total internal reflection.
 -
4. **Receiver:**
 - **Optical Detector:** A photodiode (avalanche photodiode) or a phototransistor that detects the incoming light signal and converts it back into an electrical signal.
 - **Electrical Signal Stage:** Processes and amplifies the electrical signal for further use.
 - **Demodulation:** Recovers the original information from the modulated light signal.
 -
5. **Output Device:** The electrical signal is sent to the intended output device, such as a computer, display, or network system.

**6. Describe different advantages of optical fiber communication?
Or, Describe the different benefits of fiber based communication?**

Solution:

i. Enormous potential bandwidth

- The optical carrier frequency in the range 10^{13} to 10^{16} Hz
- Coaxial cable bandwidth typically around 20 MHz over distances up to a maximum of 10 km.

ii. Small size and weight

- Optical fibers have very small diameters which are often no greater than the diameter of a human hair
- Even when such fibers are covered with protective coatings they are far smaller and much lighter than corresponding copper cables.

iii. Electrical isolation

- Optical fibers which are fabricated from glass, or sometimes a plastic polymer, are electrical insulators and therefore, unlike their metallic counterparts, they do not exhibit earth loop and interface problems.
- This property makes optical fiber transmission ideally suited for communication in electrically hazardous environments.

iv. Immunity to interference and crosstalk

- Optical fibers free from electromagnetic interference (EMI), radio-frequency interference (RFI).
- Hence the operation of an optical fiber communication system is unaffected by transmission through an electrically noisy environment

v. Signal security.

- The light from optical fibers does not radiate significantly and therefore they provide a high degree of signal security
- Unlike the situation with copper cables, a transmitted optical signal cannot be obtained from a fiber in a noninvasive manner (i.e. without drawing optical power from the fiber).
- This feature is obviously attractive for military, banking and general data transmission (i.e. computer network) applications.

vi. **Low transmission loss.**

- The development of optical fibers over the last 20 years has resulted in the production of optical fiber cables which exhibit very low attenuation or transmission loss in comparison with the best copper conductors.

vii. **Ruggedness (অমসৃণ) and flexibility.**

- Although protective coatings are essential, optical fibers may be manufactured with very high tensile strengths.
- Cable structures have been developed which have proved flexible, compact.

viii. **System reliability and ease of maintenance.**

- With fewer optical repeaters or amplifiers, system reliability is generally enhanced in comparison with conventional electrical conductor systems.

ix. **Potential low cost.**

- The glass which generally provides the optical fiber transmission medium is made from sand – not a scarce (দুর্লভ) resource.
- So, in comparison with copper conductors, optical fibers offer the potential for low-cost line communication.

7. Describe following terms with respect to optical laws?

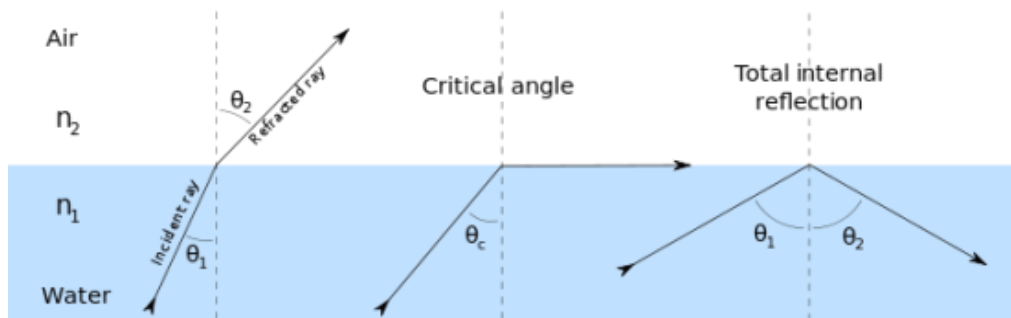
A. Critical angle and Acceptance angle.

B. Refractive Index.

Solution:

Critical angle:

The angle of incidence (আপতন), when passing from a denser medium to a lighter medium, for which the angle of refraction (প্রতিসরণ) becomes maximum (90°), meaning the refracted ray travels along the boundary, is called the critical angle of the denser medium with respect to the lighter medium.



Acceptance angle:

The acceptance angle of an optical fiber is the maximum incidence angle of a light ray which can be used for injecting light into a fiber core or waveguide.

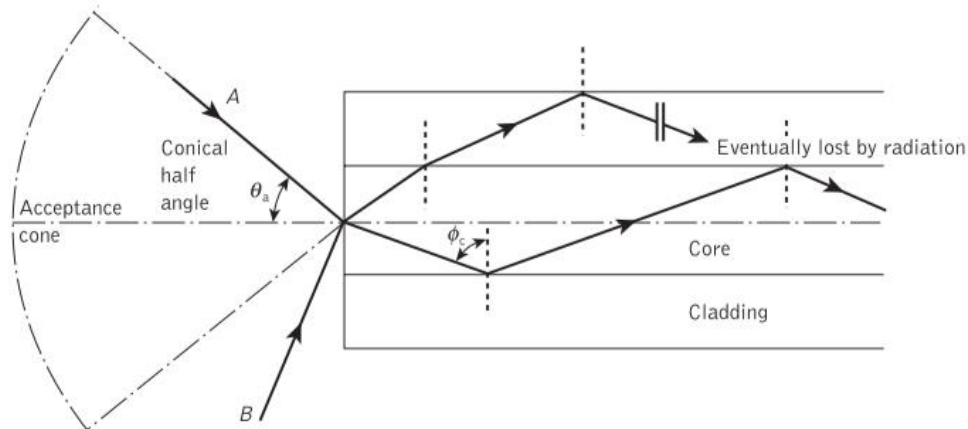
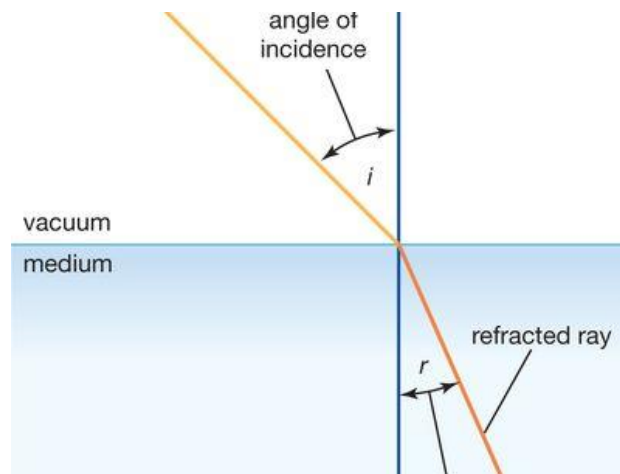


Figure 2.4 The acceptance angle θ_a when launching light into an optical fiber

Refractive Index:

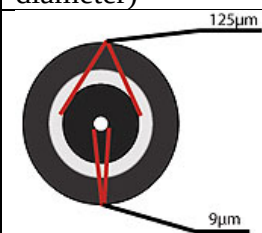
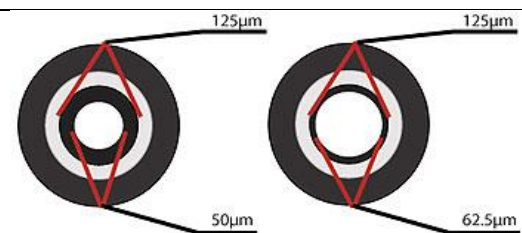
Refractive index, measure of the bending of a ray of light when passing from one medium into another.

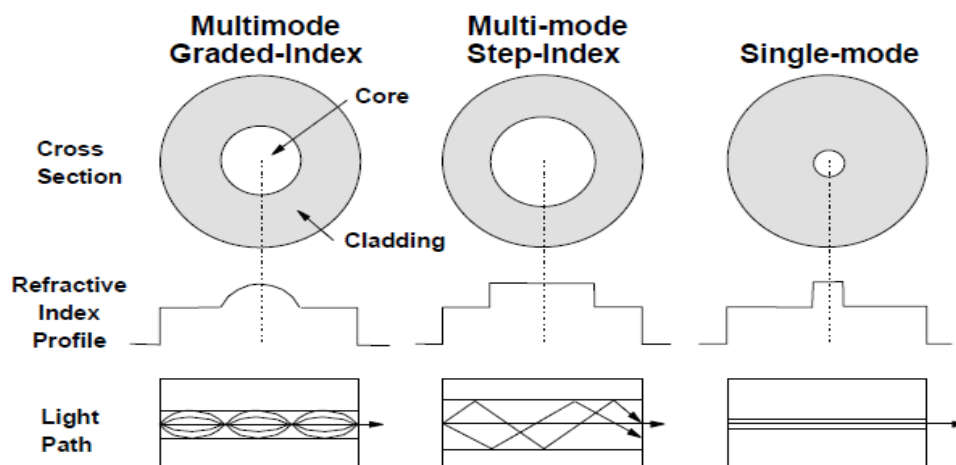
If i is the angle of incidence of a ray in vacuum and r is the angle of refraction the refractive index n is defined as the ratio of the sine of the angle of incidence to the sine of the angle of refraction; i.e., $n = \sin i / \sin r$.



8. Difference between single mode fiber optic cable and multimode fiber optic cable.

Solution:

Characteristic	Single Mode Fiber Optic Cable	Multimode Fiber Optic Cable
Definition	A fiber optic cable with a small core that allows only one mode of light to pass through.	A fiber optic cable with a larger core that allows multiple modes of light to pass through.
Core Diameter	9 microns	50 microns or 62.5 microns
Cladding Diameter	125 microns	125 microns
Number of Light Modes	Only one mode of light	Multiple modes of light
Application	Long-distance communication	Short-distance communication (LAN, data, audio/video)
Data Transmission	Lower data capacity but better for long distances	Higher data capacity but limited to shorter distances
Signal Quality over Distance	Maintains high signal quality over long distances	Signal quality reduces over long distances due to high dispersion and attenuation
Typical Use Case	Telecommunications, long-distance networks	Local Area Networks (LAN), data centers, audio/video applications
Construction Ratio	9/125 (core/cladding diameter)	50/125 or 62.5/125 (core/cladding diameter)
Figure		



9. What is an optical attenuation used for?

Solution:

- i. **Evaluating Transmission Distance:** Optical attenuation helps determine the maximum distance an optical signal can travel through a fiber before needing amplification or signal restoration.
- ii. **Comparing with Metallic Conductors:** It allows for comparing fiber optics with metallic cables; lower attenuation in fibers makes them more suitable for long-distance communication than metals.
- iii. **Signal Quality Assessment:** Attenuation measures the reduction in signal strength, which helps assess and maintain the quality of the transmitted signal.
- iv. **System Design and Optimization:** Knowing attenuation is essential for designing fiber optic systems, ensuring they are optimized for distance and performance without excessive signal loss.
- v. **Loss Mechanism Analysis:** Understanding attenuation helps identify and address various loss mechanisms in fibers, such as material absorption and scattering, to improve fiber design.
- vi. **Performance Monitoring:** Regular attenuation measurements help monitor and maintain the performance of fiber optic networks, ensuring they operate within acceptable loss limits.

10. Explain in details the intrinsic absorption loss mechanism with necessary illustration?

Solution:

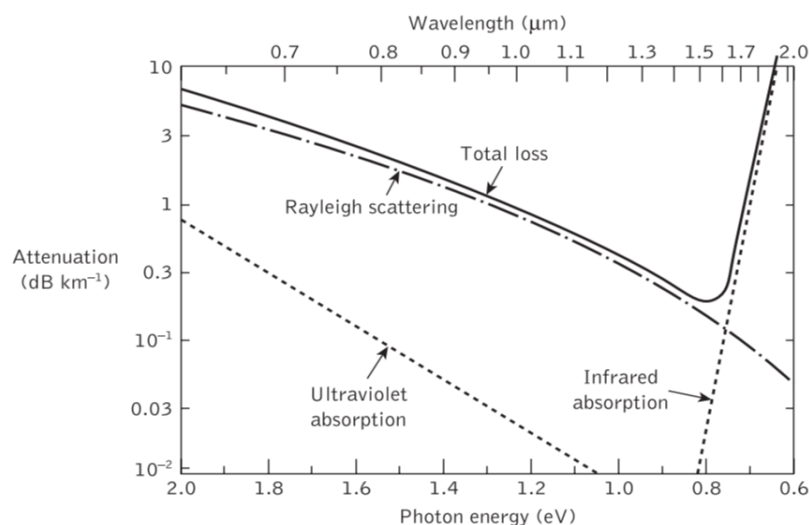


Figure 3.1 The attenuation spectra for the intrinsic loss mechanisms in pure $\text{GeO}_2\text{-SiO}_2$ glass [Ref. 5]

The graph shows the attenuation spectra for the intrinsic loss mechanisms in pure $\text{GeO}_2\text{--SiO}_2$ glass, which is commonly used in optical fibers. The attenuation is measured in dB/km (decibels per kilometer) and plotted against both photon energy (eV) and wavelength (μm).

1. Total Loss Curve:

- The total loss in the material is the sum of different attenuation mechanisms, shown by the top solid line on the graph.
- This includes losses due to Rayleigh scattering, ultraviolet (UV) absorption, and infrared absorption.

2. Rayleigh Scattering:

- Rayleigh scattering causes attenuation due to imperfections and variations in the glass structure.
- The Rayleigh scattering curve decreases as the wavelength increases, showing that this effect is more prominent at shorter wavelengths (visible and near-infrared region).

3. Ultraviolet Absorption:

- The dashed line indicates UV absorption, which increases at shorter wavelengths (toward the left side of the graph).
- This absorption occurs because the material absorbs higher-energy photons in the ultraviolet region.

4. Infrared Absorption:

- At longer wavelengths (beyond $1.6\ \mu\text{m}$), infrared absorption becomes dominant. This is due to vibrational modes in the glass structure that absorb longer-wavelength light.

5. Optical Communication Window:

- The graph shows that the lowest attenuation occurs between $0.8\ \mu\text{m}$ and $1.6\ \mu\text{m}$, which corresponds to the wavelength range used for optical fiber communications. This is because both Rayleigh scattering and infrared absorption are minimized in this range, leading to optimal performance.

In summary, the graph demonstrates that different loss mechanisms dominate at different wavelengths, with minimal loss for optical fiber communication occurring between $0.8\ \mu\text{m}$ and $1.6\ \mu\text{m}$ due to reduced scattering and absorption.